MD JARIFUL RAHMAN

Software Developer (Backend Focus)

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EDUCATION

University of Dhaka – Institute of Information Technology (IIT) Dhaka, Bangladesh
Bachelor of Science in Software Engineering (BSSE) Graduation: Oct 2026 (Coursework Completed)
CGPA: 3.34/4.00

PROFESSIONAL EXPERIENCE

Tekarsh Bangladesh Limited Oct 2025 – Mar 2026 (6 Months)
Software Developer Intern Dhaka, Bangladesh

- **Full-Stack Feature Engineering:** Contributed to a production scholarship management system (role-based education dashboard), building reusable components, refactoring core workflows, and extending analytics reporting within an Agile Scrum team.
- **Backend API Load Profiling & Stress Testing:** Ran large-scale concurrency load tests on a multi-branch restaurant management system to measure API latency and throughput, pinpointing database query bottlenecks under simulated peak traffic.
- **Automated Regression Quality Gates:** Designed and implemented automated regression smoke tests with Appium and WebdriverIO for cross-platform mobile applications, validating critical checkout and API communication flows.

TECHNICAL PROJECTS

ApexFlow – Durable Workflow Orchestration Engine GitHub | Live Demo
Python, FastAPI, PostgreSQL, Next.js 14, Web Crypto API

- Architected a code-first durable workflow engine with step-level memoization (*ctx.step*), enabling automatic recovery and instant resumption from interrupted points without re-executing completed operations.
- Engineered a high-throughput job queue on raw PostgreSQL concurrency (*FOR UPDATE SKIP LOCKED*), eliminating queue-cluster overhead, paired with client-side AES-256-GCM zero-knowledge payload encryption.
- Implemented transparent payload offloading to object storage for payloads over 250KB and built a full-stack Next.js 14 dashboard for execution timelines and worker telemetry.

InboxOrchestrator AI – Autonomous Email & Semantic Platform GitHub | Live Demo
Python, FastAPI, Supabase (pgvector), PyTorch, spaCy, Google Gemini API, Next.js 16 (React 19)

- Built a background ingestion service that syncs Gmail through incremental cursor pointers (historyId), with automated checkpointing and recovery so no messages are dropped.
- Engineered a multi-stage ML pipeline combining local classifiers, spaCy entity extraction, and 384-dimensional pgvector embeddings with the Gemini API for delta summarization, reply drafting, and task creation.
- Designed an SLA-aware thread state machine with automated escalation, noisy-vs-actionable sender analytics, database-enforced task deduplication in PostgreSQL, and 1-click Google Calendar sync.

FabricFlow – Garments Inventory Backend (RBAC) GitHub
Node.js, Express.js, PostgreSQL, Redis, Docker, Jest

- Built a high-performance backend for garments inventory, supply order tracking, and role-based access control (RBAC), with Redis-backed session authentication.
- Optimized inventory aggregation queries with tuned raw SQL (zero ORM overhead); containerized services with Docker and wrote Jest test suites.

Study Sync – Real-Time Collaborative Platform GitHub
Node.js, Express.js, Socket.io, AWS S3, MongoDB, React

- Architected a real-time collaborative hub using bidirectional WebSockets (Socket.io) for multi-room chat, presence, and live updates.
- Integrated AWS S3 with presigned URLs for secure, direct academic document uploads, offloading file traffic from the server.

CodeBlog API – Modular RESTful Blogging Engine GitHub
Java, Spring Boot, PostgreSQL, JWT

- Engineered a modular RESTful blogging API with a clean package-by-feature architecture separating authentication, article publication, and threaded comments.
- Implemented stateless JWT route authorization, custom exception handling, and transactional database integrity, following SOLID design principles.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, SQL
Backend: FastAPI, Node.js, Express.js, Spring Boot
Databases & Cloud: PostgreSQL, MongoDB, AWS S3
Frontend: Next.js, React, Tailwind CSS, HTML/CSS
DevOps & Tools: Docker, Linux, Git, GitHub, Postman, Nginx